CREW SELECTION & TRAINING



On Feb. 27, 2024, India got its four designated 'Gaganyatris'—all Group Captains in the IAF

- Shubhanshu Shukla
- Prasanth Nair
- Ajit Krishnan
- Angad Pratap

Only three will be part of the first crewed mission

STATUS
Physical, psychological and generic training completed; Shukla travels to the ISS on Axiom-4 mission

CREW RECOVERY OPERATIONS

Recovery Team
ISRO + Indian Navy

Post-landing in sea

- Divers retrieve crew and module
- Medical checkups onboard recovery vessels
- Transport to mainland for debriefing

STATUS:
Validation of CS deceleration, flotation system, parachute drop tests completed



Narayanan. "Within the minimum space, we aim to provide maximum comfort and utility so that it will be cost-effective."

India is keen to use its own upcoming space station for many cutting-edge experiments, some of which Shukla is already conducting in his two-week sojourn aboard the ISS. These include examining human interaction with electronic displays in microgravity and identifying any musculoskeletal dysfunctions in Zero-G as well as the effects of metabolic supplements. There are many studies designed for Shukla to analyse the impact of microgravity on cultivating food crops like moong dal, methi sprouts and on edible microalgae. There is also an indepth experiment on tardigrades—microscopic



"Gaganyaan is a national project that not only involves ISRO but also the Indian navy and air force, apart from a whole host of industries in the public and private sector"

—DR V. NARAYANAN
Chairman, ISRO

aquatic animals—and their functioning in such harsh conditions. All these experiments have been devised by key scientific institutions across the country and many more would be done when BAS-1 comes up. The idea is to prepare for the exploitation of lunar resources and possibly Mars in the near future.

JOINING THE TITANS

All these programmes segue neatly into the long-term strategy of India becoming a major player in the lunar race. After ISRO successfully demonstrated its capability of landing a spacecraft on the moon surface and operating a rover in August 2023, the Modi government sanctioned the ambitious Chandrayaan-4 project in September 2024. It involves the development of technology to land a spacecraft on the moon, collect rock samples and then fly them back to Earth for analysis. Estimated to cost upward of Rs 2,000 crore, Chandrayaan-4 is expected to be launched in 2027-28, and will demonstrate foundational technological capabilities for an Indian astronaut landing on the moon by 2040 and returning safely to the earth. It will be followed by Chandrayaan-5, a joint mission with Japan that will have a